

*MA.6.DP.1.6***Benchmark**

MA.6.DP.1.6 **Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.**

Benchmark Clarifications:

Clarification 1: Instruction includes choosing the measure of center or measure of variation depending on the scenario.

Clarification 2: The measures of center are limited to mean and median. The measures of variation are limited to range and interquartile range.

Clarification 3: Numerical data is limited to positive rational numbers.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.2.3
- MA.6.AR.1.2
- MA.6.AR.2.3
- MA.7.DP.1.1, MA.7.DP.1.2

Terms from the K-12 Glossary

- Data
- Interquartile Range
- Mean
- Measures of Center
- Measures of Variability
- Median
- Range (of a data set)

Vertical Alignment**Previous Benchmarks**

- MA.5.DP.1.2

Next Benchmarks

- MA.8.DP.1.1
- MA.912.DP.1.1

Purpose and Instructional Strategies

In previous courses, students interpreted numerical data by determining the mean, median, mode and range when the data involved whole-number values and was represented with tables or line plots. In grade 6 accelerated, student understanding is extended to determine the measures of center and variation when the data points are positive rational numbers. Students are also expected to use their understanding of arithmetic to describe how the addition or removal of data values will impact the measures of center and variation within real-world scenarios. Students will also determine an appropriate measure of center or measure of variation to summarize data and use them to make comparisons when given two numerical or graphical representations of data. In future courses, students are introduced to numerical bivariate data and depict it with line graphs and lines of fit. Students will also select an appropriate method to represent both univariate and bivariate numerical data and interpret the different components in the display.

- Instruction includes student understanding that the choice between using the median or the mean as the measure of center or using the interquartile range or range as the measure of variation depends on the purpose.

- For example, in some cases it is important to include the influence of outliers in the description of a population and the mean might be preferable as the measure of center and the range might be preferable as the measure of variation. If the influence of outliers is meant to be ignored, the median and interquartile range may be more preferable.
- Students should be provided with data sets and asked to find the measures of center and of variability. Students should explore and look for patterns when a data value is removed, or a new data value is added that meets the following conditions (*MTR.5.1*):
 - The value is larger than the measure of the center.
 - The value is equal to the measure of the center.
 - The value is less than the measure of the center.
 - The value is greater or less than any value of the original data set.
 - The value is within the range of the original data set.
- Instruction includes the use of technology to help with connecting the measures of center and variation to the visual representation on a box plot. This can help students understand what the value of the median, range and interquartile range represent within a context. If a value is added or removed from a data set, students can create a second box plot based on the new data set to help determine and describe the impact of the change (*MTR.2.1*).

Common Misconceptions or Errors

- Students may incorrectly categorize the data measures (mean, median, range, interquartile range) as measures of center or measures of variability. It can be helpful to frequently refer back to the purpose of a type of measure and how it relates to the center of the data or the variation of the data.
- Students may incorrectly try to determine the change by calculating the mean, median, range or interquartile range for the changed data set instead of calculating for both the initial and the changed sets before comparing.

Strategies to Support Tiered Instruction

- Instruction includes having students calculate the measures of center (mean and median) or the measures of variation (range and interquartile range) for the initial data sets and changed data sets before comparing the impact of an outlier or an additional data point.
- Instruction includes co-creating a graphic organizer for measures of center (mean and median) and measures of variation (range and interquartile range), and include the generalized impact of outliers on each.

Instructional Tasks

Instructional Task 1 (MTR.5.1, MTR.6.1, MTR.7.1)

Mr. Biggs and his students shared the number of hours they played video games last week and Mr. Biggs recorded it in the table below.

2.5	12	0	28	6.75	21.25	1.2	16
7.33	1.5	4	15	7	2.5	16	0

- Part A. Find a measure of center and variability to represent this scenario. Explain why you chose these measures.
- Part B. Mr. Biggs removed the 4 from the data set since this represented the number of hours that he played video games. Describe the impact this removal has on the measures of center and measures of variability of the data set.
- Part C. Mario is a new student in Mr. Biggs's class. If the number of hours Mario played video games caused the measure of center to increase, how long could Mario have played video games last week? Explain your reasoning.

Instructional Items

Instructional Item 1

Christian took his temperature, in degrees Fahrenheit (°F), every day for a week. His temperatures are shown in the table below:

Day	Temperature (°F)
Sunday	98.7
Monday	99.1
Tuesday	98.2
Wednesday	97.9
Thursday	98.1
Friday	98.8
Saturday	99.2
Sunday	

- Part A. Which measure of center and variation could be used to summarize the data? Explain your choice.
- Part B. He took his temperature on the following Sunday, and it was 103.2°F. How does this impact the measures of center and variation?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*